

ORIGINAL RESEARCH

Measuring Physiological Stress in Mammals: How Metabolism Influences Fecal Glucocorticoid Levels

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Received: 5 January 2026 | **Revised:** 30 July 2026 | **Accepted:** 3 September 2026

Associate Editor: Desire Dalton

Keywords: allometry | energetics | fecal glucocorticoids | mammals | metabolic theory | scaling | steroid hormones | stress physiology

ABSTRACT

The concept of physiological stress is widely used but poorly understood. In mammals, physiological stress is often assessed noninvasively based on the concentration of glucocorticoid hormone (i.e., cortisol, corticosterone) metabolites in feces (i.e., fecal glucocorticoids—FGCs). Yet, it remains unclear if FGC levels are measuring stress and how such levels vary across species. Previous work has shown that much of the variation in plasma cortisol levels in mammals is explained by differences in species' metabolic rates (i.e., rates of energy use), indicating that differences in plasma cortisol levels may simply be reflecting differences in metabolic rate rather than differences in physiological stress levels. Here we examine whether variation in FGCs is similarly explained by differences in species' metabolic rates among mammals. As predicted, data from 77 mammal species show that FGCs increase in proportion to mass-specific metabolic rate (MSMR) and similarly scale to the $-1/4$ power of body mass. Most variation in baseline FGC levels across species was explained by differences in MSMR or body mass. Furthermore, after exposure to stressors, elevated FGC levels increased in proportion to baseline levels. Taken together, these results suggest that increases in FGC levels in response to a perceived threat or perturbation are largely measuring changes in rates of energy use (i.e., metabolic rate).

1 | Introduction

Biologists are increasingly interested in characterizing stress, but doing so is fraught with challenges. Stress has been defined as a real or perceived threat to a species' well-being that leads to short term changes in a broad suite of physiological (e.g., heart rate, blood pressure) or psychological (e.g., mental acuity) factors that may increase performance (Levine 2005; McEwen 2007). Thus, the stress response is often described as a temporary disruption of homeostasis (Koolhaas et al. 2011). However, a more specific operational definition that captures the underlying physiology has proven elusive given the nonspecific nature of the definition of stress. For this reason, Randy Nelson once described stress as a “notoriously ethereal concept” (Nelson 2005). Nevertheless, biologists continue to use various measures to characterize stress

levels. In the case of physiological stress, which for vertebrates involves the interplay of the autonomic nervous system (sympathetic and parasympathetic) and the HPA (hypothalamus–pituitary–adrenal) axis, glucocorticoid hormone levels (cortisol and/or corticosterone) are typically used.

Glucocorticoid hormones (“GCs”) are used across subdisciplines to measure stress in vertebrates (Romero 2004). These signaling molecules are produced by the HPA axis as part of a feedback loop that is thought to regulate at least some functions involved in the stress response (Nicolaidis et al. 2014), including carbohydrate metabolism and the maintenance of blood glucose levels (Kuo et al. 2015). Ecologists use GCs to assess the response of species to a wide range of potential stressors, from environmental change and anthropogenic disturbance to biotic interactions

(Dantzer et al. 2014; Romero and Beattie 2022). Physiologists, biomedical scientists, and social scientists use GC levels to assess the physiological or emotional response of individuals to a range of challenging circumstances (e.g., occupational stress; Lee et al. 2015). Yet, it remains unclear to what extent GCs represent a general measure of stress (MacDougall-Shackleton et al. 2019).

To assess whether GCs represent a general measure of stress, we must better understand the factors that contribute to variation in GC levels. In particular, we lack sufficient understanding of the relative importance of factors that may contribute to heterogeneity in GC levels across species. In part, this is because most studies of physiological stress using GCs are focused on individuals or populations. They aim to compare GC hormone concentrations in blood, tissue, or feces prior to a stressor being applied (i.e., baseline levels) and shortly after exposure to this stressor (i.e., elevated levels) among these groups. Thus, these studies do not address (and are not intended to address) the substantial interspecific variation in GC levels across species. For example, plasma cortisol levels vary by more than 50-fold in both mammals and birds (Haase et al. 2016; Rubalcaba and Jimeno 2022a).

Research by Haase et al. (2016) potentially provides a point of departure for comparing GC levels across species. This study has shown that plasma cortisol increases linearly with mass-specific metabolic rate among cortisol-dominant mammals and decreases nonlinearly with body mass in much the same way as mass-specific metabolic rate. Further examining the relationship between GCs and metabolic rate could be particularly useful for cross-species comparisons since metabolic rate varies somewhat predictably with body size and temperature across vertebrates (Gillooly et al. 2001). This is particularly true for measures of GC levels using fecal glucocorticoid metabolites, which are increasingly the measure of choice for glucocorticoids because they are non-invasive measures of adrenocortical activity (Touma and Palme 2005; Sheriff et al. 2010).

Here we build on the study of Haase et al. (2016) by examining the relationship of fecal glucocorticoid metabolites to metabolic rate and body mass in mammals. We first evaluate whether the Haase et al.'s (2016) model that relates plasma cortisol to mass-specific metabolic rate and body mass in mammals applies to FGCs such that:

$$[\text{Glucocorticoid metabolites}] = a \frac{B}{M} = ab M^{-0.25} \quad (1)$$

where a is a constant that relates glucocorticoid metabolite concentration to metabolic energy flux per gram of tissue ($\text{ng GC (g feces)}^{-1} \text{mW}^{-1} \text{(g tissue)}^{-1}$), b is a normalization constant that describes the mass-independent rate of energy flux per gram of tissue ($\text{mW g}^{0.75}$), M is body mass (grams), and B/M is mass-specific metabolic rate (mW g^{-1}). Equation (1) yields two testable predictions with respect to baseline FGCs: (i) a log–log plot of FGC versus MSMR should yield a slope of 1, and (ii) a log–log plot of FGC versus body mass should yield a slope of -0.25 . After testing these predictions, we then examine the relationship between FGC levels after exposure to a stressor (i.e., elevated

levels) to baseline levels. Last, we compare the relationships of FGCs to MSMR and body mass to those reported for plasma cortisol in cortisol-dominant mammals. We examine these aims using extensive data from a diversity of terrestrial and volant placental mammals.

2 | Methods

We compiled published data on baseline and elevated fecal glucocorticoid metabolites (FGCs) across 77 mammal species from 31 families (Supporting Information, Appendix 3, Tables S2 and S3). When multiple values were presented for either baseline or elevated values, including measures of both males and females, we averaged the values. When multiple elevated FGC experiments were reported, we used those that produced the higher values. We used the reported body masses of the study subjects from the original studies when available, but otherwise used adult body masses taken from AnAge (de Magalhães et al. 2024) or other independent sources when these were unavailable on AnAge (Supporting Information, Appendix 3, Tables S2 and S3). Similarly, mass-specific resting metabolic rate values were taken from AnAge (de Magalhães et al. 2024) or other independent sources if these were unavailable on AnAge (Supporting Information, Appendix 3, Tables S2 and S3) since metabolic rates were not measured in the original studies. Finally, we collected data on the life stage (i.e., adult, juvenile), sex, reproductive status, assay type, stressor type, number of individuals measured (i.e., “ n ”), and whether subjects were captive or wild. Note, though, that nearly all subjects were adults and reproductive status was generally not indicated. See Supporting Information, Appendix 4 for additional details.

Data on FGC levels were categorized based on assay type to account for possible differences in the sensitivity and cross-reactivity of different assay types (Möstl et al. 2002; Palme 2019). We categorized assays into two categories: (1) those using assays designed to measure corticosterone metabolites and (2) those designed to measure cortisol metabolites. Data from each category were analyzed and presented independently to avoid issues of data nonindependence since about 20% of the species considered here had measures using both categories of assays. In these cases, the extent to which each set of measures for each species is nonindependent cannot be determined since the relative degree of cross-reactivity for any particular set of assays is unknown. New World primates were excluded from analyses because they are known to have GC values 50–100 times higher than other mammals presumably to overcome GC resistance at the organ level that is attributed to factors such as low receptor affinity and/or low concentrations of cellular steroid receptors (Klosterman et al. 1986; Brandon et al. 1989; Westberry et al. 2006).

The first category of assay included studies designed to measure corticosterone or its metabolites. This category consisted of five commonly used EIA or RIA assays, the vast majority of which were either the Corticosterone EIA assay (Munro, CJM06), which originated in 2006, or the I^{125} RIA available as a kit from ICN Biomedicals. The second category of assays, those generally used to measure cortisol or its metabolites, consisted mostly of studies using one of four commonly used and

validated EIA assays (Munro, R4866, Möstl et al. 2002, Palme and Mostl 1997, Touma et al. 2003). The small number of remaining assay types in this category were similar in nature and comprised of commercially available kits (e.g., Enzo Life Sciences Cortisol Kit).

We tested the relationships between baseline FGCs and MSMR, and between baseline and elevated levels, using phylogenetic generalized least squares (PGLS) regression to account for evolutionary relatedness among species. To perform these analyses, we first constructed a mammal phylogeny of our taxa (Supporting Information, Appendix 1) from the Open Tree of Life (Cranston et al. 2024) using the R package APE (Paradis and Schliep 2019). For each regression analysis, the phylogenetic tree was trimmed to match the subset of species considered, and a covariation matrix was constructed assuming a Brownian motion model of evolution. Off-diagonal elements of the covariation matrix were then scaled to the phylogenetic signal (λ) of the response (Revell and Harmon 2022). The magnitude of phylogenetic signal varied across models but was sufficiently large ($\lambda > 0.3$) to warrant meaningful phylogenetic correction in most cases (Table 1). For consistency, all models were therefore analyzed using PGLS with the R package nlme (Pinheiro

et al. 2025), though for those analyses with weak phylogenetic signal the model approaches that of an ordinary least squares regression.

We accounted for the possible effects of assay type, sex, captive vs. wild subjects, and stressor type (ACTH vs. all other stressor types) in our models by considering them as categorical variables, fitting models with these variables, and comparing model fits based on Bayesian information criterion (BIC). Statistics regarding the fits of the best performing models were described with a R^2 value based on a likelihood approach using the rr2 package (Ives and Li 2018). All analyses were done in R version 4.4.2 (R Core Team 2025).

3 | Results

Data largely support the predicted relationship between baseline FGCs and metabolic rate (Equation 1). Baseline FGC concentrations increase with MSMR for both cortisol- and corticosterone-specific assays (Figure 1A,B) in a manner that is not significantly different from unity based on the 95% confidence intervals of the fitted slopes (est: 1.53 and 1.18; 95% CI: [0.86, 2.21] and [0.70,

TABLE 1 | Phylogenetic generalized least squares regression statistics for relationships shown in Figures 1–3.

Model	Intercept (95% CI)	Slope (95% CI)	Lik. R^2	p	λ	n
Cortisol-based assay						
Baseline FGC versus MSMR	3.95 (2.91, 5.00)	1.53 (0.86, 2.21)	0.42	<0.001	0.36	29
Baseline FGC versus mass	7.97 (6.17, 9.78)	−0.34 (−0.46, −0.22)	0.47	<0.001	0.66	52
Elevated FGC versus baseline FGC	1.42 (0.65, 2.19)	0.97 (0.82, 1.13)	0.78	<0.001	0.04	43
Corticosterone-based assay						
Baseline FGC versus MSMR	4.14 (3.24, 5.04)	1.18 (0.7, 1.66)	0.67	<0.001	0.47	23
Baseline FGC versus mass	7.20 (5.80, 8.59)	−0.29 (−0.40, −0.18)	0.55	<0.001	0.58	44
Elevated FGC versus baseline FGC	2.41 (0.97, 3.85)	0.83 (0.58, 1.09)	0.71	<0.001	0.42	29

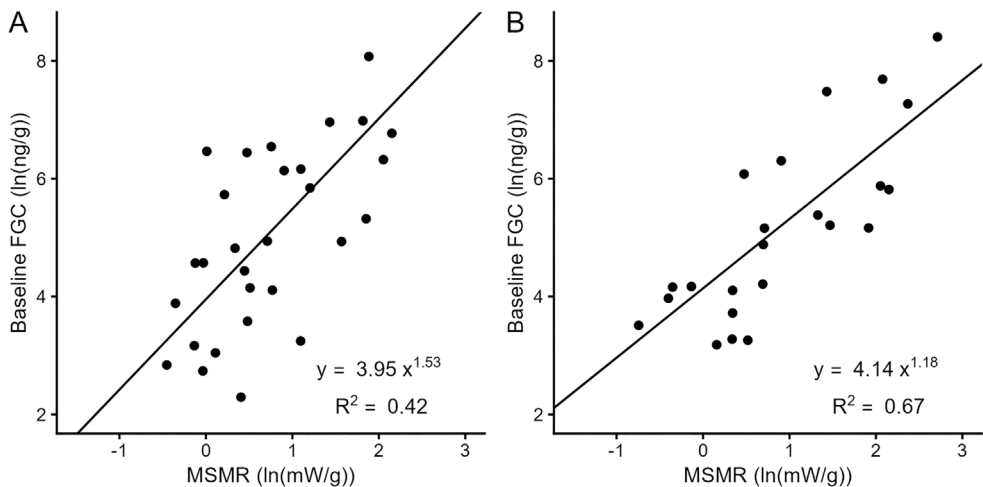


FIGURE 1 | (A and B) Relationships between baseline fecal glucocorticoid metabolite concentrations (i.e., FGCs) and mass-specific metabolic rate for data using cortisol-based assays (A, $n = 29$) and corticosterone-based assays (B, $n = 23$). Data are natural log-transformed, and statistics are presented in Table 1.

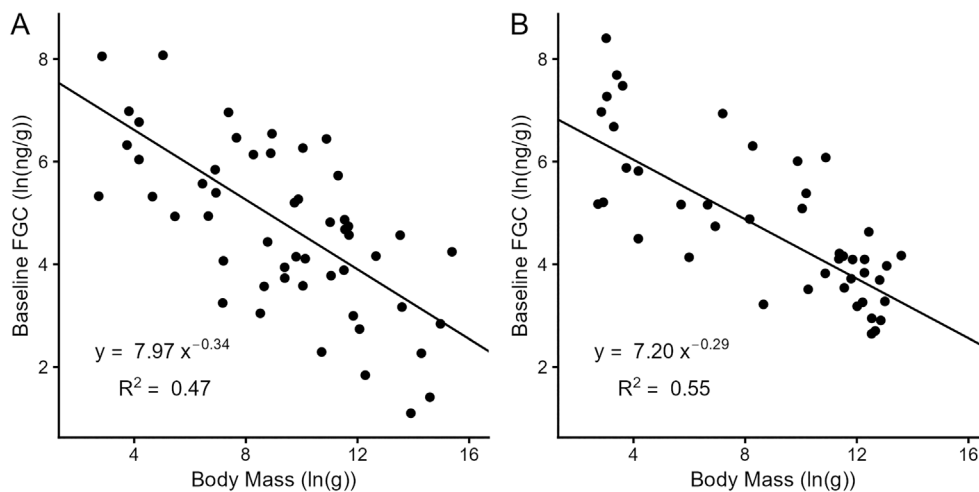


FIGURE 2 | (A and B) Relationships between baseline fecal glucocorticoid metabolite concentrations (i.e., FGCs) and body mass for data using cortisol-based assays (A, $n = 52$) and corticosterone-based assays (B, $n = 44$). Data are natural log-transformed, and statistics are presented in Table 1.

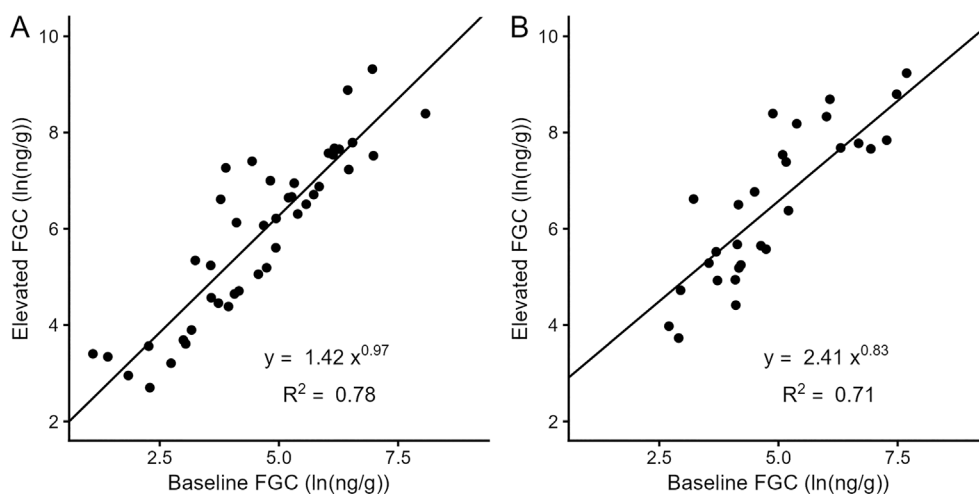


FIGURE 3 | (A and B) Relationships between elevated fecal glucocorticoid metabolite concentrations (i.e., FGCs) and baseline FGCs for data using cortisol-based assays (A, $n = 43$) and corticosterone-based assays (B, $n = 29$). Data are natural log-transformed, and statistics are presented in Table 1.

1.66], respectively; Table 1). Including assay type, sex, captive versus wild subjects, and stressor type as categorical variables did not improve models based on BIC criteria in most cases and was therefore omitted from all models (Supporting Information, Appendix 2).

Data also largely support the predicted relationship between baseline FGCs and body mass (Equation 1). Log-log plots of baseline FGCs versus body mass yield slopes (-0.34 , -0.29) that are not significantly different from the predicted value of -0.25 for both cortisol- and corticosterone-specific assays (Figure 2A,B, 95% CI: $[-0.46, -0.22]$ and $[-0.40, -0.18]$, respectively). Thus, on average, baseline FGCs decrease by approximately 50% for every order of magnitude increase in body mass.

Both MSMR and body mass explain a substantial portion of the variation in baseline FGCs across species. MSMR explains 42% and 67% of the variation in baseline FGCs, for cortisol and corticosterone-specific assays, respectively, based on likelihood

R^2 (Figure 1A,B; $n = 29$, $n = 23$). Similarly, body mass explains 47% and 55% of the variation in baseline FGCs (Figure 2A,B). Note, too, that the relationships of FGCs to MSMR and body mass were similar in slope and intercept for both assay categories. This suggests that cortisol- and corticosterone-specific assays are similarly measuring total glucocorticoid metabolites in mammalian feces, and that total glucocorticoid levels are similar for a given body mass or MSMR regardless of the dominant hormone.

Elevated FGC concentrations increase linearly with baseline FGCs independent of the type of stressor applied (Figure 3A,B). For both assay categories, the observed slopes (0.97 and 0.83) are not significantly different than unity (95% CI: $[0.82-1.13]$, $[0.58-1.09]$). Moreover, in both cases, the increase from baseline to elevated FGC levels was similar. Elevated FGC levels were 3.4-fold higher than baseline at mean values (about 320 ng/g baseline). These results suggest the magnitude of the stress response may be somewhat fixed across species. Baseline FGCs explain 78% and 71% of the variation in elevated levels, respectively, irrespective of the stressor type (Figure 3A,B).

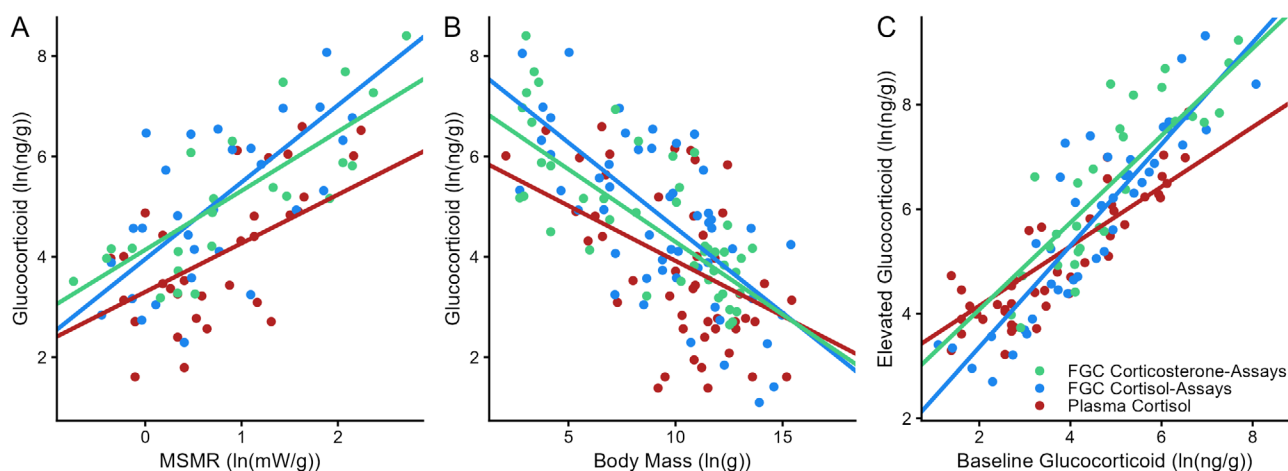


FIGURE 4 | (A–C) Panels A and B compare the relationships of baseline FGCs versus baseline plasma cortisol to mass-specific metabolic rate (A) and to body mass (B). Panel C compares the relationships of baseline to elevated plasma cortisol versus baseline to elevated FGC values. Data are natural log-transformed.

Our results (Figures 1–3) using FGCs are consistent with those reported for plasma cortisol in cortisol-dominant mammals from Haase et al. (2016) (Figure 4A–C). First, our analyses of baseline FGCs and Haase et al.'s (2016) analysis of plasma cortisol found reasonably strong, linear correlations with MSMR (Figure 4A). Second, the slopes of the relationships between FGC and body mass are quite similar to those for plasma cortisol (Figure 4B). Third, both our analyses and those of Haase et al. (2016) found strong, positive relationships between baseline and elevated levels (Figure 4C). However, Haase et al. found elevated levels increased sub-linearly (slope = 0.57) with baseline levels in log–log plots, whereas we observed linear increases.

4 | Discussion

The relationships shown between FGCs and metabolic rate (Figure 1, Table 1), and those between FGCs and body mass (Figure 2, Table 1), indicate that FGC levels are tightly linked to rates of energy use across species. Thus, changes in FGC levels after exposure to a stressor are largely capturing the energetic response of species to these disturbances as opposed to the overall stress response. We are not aware of any previous research showing that variation in FGC levels across species is related to differences in rates of energy use or body mass. Still, this conclusion is broadly consistent with research that recognizes the stress response, as inferred by GC levels measured in blood or tissue, has a strong metabolic component that is characterized by increases in rates of energy use (Koolhaas et al. 2011; Buwalda et al. 2012; Haase et al. 2016; Francis et al. 2018; MacDougall-Shackleton et al. 2019; Gormally and Romero 2020; Romero and Beattie 2022). In particular, one previous study shows the same predicted body mass and temperature dependencies between plasma glucocorticoid levels, metabolic rate, and body mass in mammals (Haase et al. 2016), and another shows weaker support for these same predicted dependencies across vertebrates (Francis et al. 2018). To that end, some have suggested that measures of GC hormone levels are best viewed as proxy measures of metabolic rate as opposed to measures of stress (Jimeno et al. 2018; Jimeno and Verhulst 2023). Thus, one might expect that biotic or abiotic factors that lead to increases in metabolic

rate may also lead to increases in FGC levels. Previous research has shown that increases in the activity levels of study subjects that increase metabolic rate may lead to increases in plasma GC levels (Koolhaas et al. 2011; Buwalda et al. 2012; Rubalcaba and Jimeno 2022b). Other research in birds has shown that increases in heat dissipation rate with decreases in environmental temperature may lead to increases in plasma GC levels since metabolic rate must increase to maintain a constant body temperature—particularly at temperatures below a species' thermal neutral zone (Rubalcaba and Jimeno 2022a; see also Jessop et al. 2016). However, that is not to say that endotherms living in colder climates should necessarily have higher metabolic rates and higher GC levels. Research in birds and mammals has shown that field metabolic rates do not vary predictably with environmental temperature (Anderson and Jetz 2005).

The observation that increases in FGCs after exposure to a stressor increase in proportion to baseline FGCs (Figure 3) points to further links between FGCs and metabolic rate. Specifically, they suggest that the fold-increase in FGCs, often interpreted as the magnitude of the stress response, may be related to metabolic scope. Elevated FGC levels are about 3.4-fold higher than baseline levels on average (Figure 3A,B), which is similar to the 3.5-fold difference between field and maximal metabolic rates in mammals (Savage et al. 2004) (a 22 kg animal). This difference between elevated and baseline values in FGCs is nearly identical to that observed for plasma cortisol in mammals by Haase et al. (2016). And, interestingly, this holds despite variation in stressor types across species. This suggests that differences in the magnitude of the response to different stressor types (ACTH injection, capture, etc.) are relatively small relative to differences in baseline FGC values. This is somewhat surprising since one might expect that injections of adrenocorticotrophic hormone (ACTH), which directly signal the adrenal cortex to produce glucocorticoids, would have a more pronounced effect on FGC levels than other, potentially milder stressors. However, this is difficult to assess since the magnitude or intensity of a particular stressor type (including the quantity of ACTH administered) varies considerably. For example, “animal capture” is a stressor type that may elicit a very different response depending on the difficulty of capture.

There is a growing recognition of the link between GCs and rates of energy use (Koolhaas et al. 2011; Buwalda et al. 2012; Haase et al. 2016; Francis et al. 2018; Jimeno et al. 2018; MacDougall-Shackleton et al. 2019; Gormally and Romero 2020; Romero and Beattie 2022; Jimeno and Verhulst 2023), as discussed above, but many studies of physiological stress still do not account for differences in body mass or metabolic rate in comparing GC levels across individuals or species. This is particularly true for studies using FGCs as a measure of glucocorticoid levels. In the studies compiled for the analyses shown here, no studies normalized or corrected for differences in body mass or metabolic rate. In the absence of such a correction, it is not at all clear what additional information is gained by measuring GC levels as opposed to metabolic rate or heart rate in gauging a species' response to threat or perturbation. At the very least, the results shown here point to the need to account for the nonlinear effects of body mass on metabolic rate when using GCs to infer stress levels across species or populations. If not, one could erroneously conclude that a smaller-bodied organism is more stressed on account of it having a higher mass-specific metabolic rate. Simply put, high levels of FGCs do not necessarily point to higher levels of stress or poor health (Romero 2004; Romero and Beattie 2022).

Further understanding the metabolic component of stress, or the extent to which a species response to threat or perturbation involves increases in rates of energy use, may be helpful in assessing how the cumulative effects of stress negatively impact wellness or longevity. Quantifying these cumulative effects is currently quite difficult based on hormone concentration measurements. Instead, we rely on subjective descriptors like "chronic" or "acute" to describe the magnitude or duration of stress episodes, or "good" (i.e., eustress) and "bad" stress (i.e., distress) to describe if levels are helpful or harmful (Selye 1975). However, there is some evidence that the metabolic component of stress (i.e., increases in rates of energy use) may be largely responsible for the detrimental effects of chronic stress. The negative effect of high GC levels on a range of organismal function, from cognition to immune responses (Sapolsky 1999; Dhabhar 2014), are often attributed to GCs' positive effect on oxidative damage (McIntosh and Sapolsky 1996; Sapolsky 2004) via free radical production—an inherently metabolic process (Costantini et al. 2011). In other words, the mechanisms by which high GC levels may cause "wear and tear" are the same as those invoked by more explicitly energetics-based measures of stress (i.e., oxidative stress) and energetics-based theories of aging (e.g., the Free Radical Theory of Aging). This may explain why free radical production (Kawanishi and Oikawa 2004), physiological stress (Epel et al. 2004), and resting metabolic rates (Gillooly and Khazan 2024) have all been linked to telomere shortening rates.

Still, we recognize that a better understanding of the relationship between FGC levels and metabolic rate represents only a step toward developing a general framework for comparing stress levels across species. The action of GCs cannot be fully understood without considering the function and density of binding proteins and GC receptors (Romero and Beattie 2022). From a practical perspective, a higher degree of standardization in measures would also be helpful in cross-species comparisons. For example, baseline FGC values are not taken under standardized conditions, as with basal measures of

metabolic rate where all animals are at rest. They simply represent the FGC values prior to the application of the stressor. Similarly, elevated stress levels do not necessarily reflect peak values. Some of the unexplained variation in our plots likely reflects this lack of standardization across studies. Similarly, some of the unexplained variation may be because the body masses in most cases, and the metabolic rates in all cases, were not those of the study subjects. Furthermore, most reported values of FGCs reported here for a species represent averages of multiple adult individuals that presumably vary in body mass. While this intraspecific variation in adult mass is unlikely to have much of an effect on the broad-scale interspecific relationships shown here, such variation may be important to consider in single species studies.

Nevertheless, our results showing that FGC levels are linked to rates of energy use in mammals raise important issues regarding their use and interpretation as general measures of physiological stress. To begin, they suggest that one should expect baseline GC levels to vary predictably with body mass and body temperature in much the same way as mass-specific metabolic rate. This provides a point of departure for cross-species comparisons. But more broadly, they suggest that the stress response itself (as inferred from changes in GC levels) is largely an energetic response. Our results showing that the fold increase in FGC levels after exposure to a stressor corresponds to metabolic scope is consistent with this interpretation. Together these results point to the need to develop a more quantitative approach to stress that considers the physiological response to threat or perturbation as a dynamic, energy-mediated rate process. Such an approach could potentially lead to predictions regarding how the time scale of the stress response varies across species. For example, one might reasonably expect that it takes predictably longer for an elephant to reach its peak or maximum level of stress after exposure to a stressor as compared to a mouse on account of the elephant's lower mass-specific metabolic rate. At present, we lack a framework for comparing the time scale of the stress response across species. Moreover, it could enhance our ability to assess the cumulative effects of recurrent episodes of stress on a range of life history processes tied to energetics (e.g., lifespan, reproduction, growth) that vary predictably with mass-specific metabolic rate. If chronically high stress is reflected in a chronically high mass-specific metabolic rate, then one might expect to see proportional increases in any number of rates that covary with mass-specific metabolic rate (e.g., growth rates, reproductive rates, mortality rates). That is, perhaps chronic stress accelerates not just the rate of aging, but all energetics-based rates of a species' life history.

Author Contributions

K.H. and J.F.G. led the writing of the manuscript. J.F.G. conceived ideas and guided data collection. A.L. compiled and collated data and performed preliminary analysis assisted by J.F.G., K.H. led the analysis assisted by A.R.W. and J.F.G. All authors contributed to drafts and gave final approval for publication.

Acknowledgments

We thank A. Longo and D. Julian for meaningful discussions on the manuscript, Z. Colón-Piñero for her help with troubleshooting our

script, A. Lavelle and D. A. Bowers for reviews and comments on the manuscript, and members of the PEERs ecology seminar for their constructive comments in the early stages of the project.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Appendix 1:** Phylogenies used in analyses. **Appendix 2:** Table S1 showing model selection statistics. **Table S1:** Model selection table using Bayesian Information Criterion. Method refers to assay type (EIA, RIA, HPLC), "Captive" compares captive to wild animals, "Sex" compares females, males or both, and "Stressor" compares studies using ACTH injections to those using other types of stressors. **Appendix 3:** Tables S2 and S3 provide the data used in analyses, followed by the references cited in each table. **Table S2:** Fecal glucocorticoid metabolite concentration (ng/g) data for mammals measured using cortisol-specific assays. Data include body mass (g), mass-specific metabolic rate (MSMR) (mW/g), assay type, stressor type, sex, number of sampled individuals, and captive or wild individuals. **Table S3:** Fecal glucocorticoid metabolite concentration (ng/g) data for mammals measured using corticosterone-specific assays. Data include body mass (g), mass-specific metabolic rate (MSMR) (mW/g), assay type, stressor type, sex, number of sampled individuals, and captive or wild individuals. **Appendix 4:** Additional information on data selection criteria.